Lecture 4

Simple, solvable and nilpotent groups

Objectives. *Lecture 3 found orders, such as pq and 12, at which no group is simple. This lecture proves that A_5 is simple, and shows how every finite group is built from simple groups: a composition series breaks it into simple factors, and by the Jordan–Hölder theorem the factors do not depend on the series. The groups built from abelian pieces are the solvable groups; S_4 is one, and S_n for $n \geq 5$ is not. The commutator subgroup goes on to Lecture 6, and solvability to Lecture 9.*

By the end of this lecture you will be able to:

- Decide whether a class of S_n inside A_n is one class of A_n or two, and prove that A_5 is simple.
- Prove that every finite group has a composition series, and that any two have the same factors, up to order and isomorphism.
- Compute the commutator subgroup and the derived series of a small group, and prove that S_n is solvable exactly when $n \leq 4$.
- Prove that p-groups are nilpotent and nilpotent groups solvable, and show that S_3 is solvable but not nilpotent.

4.1 Simple groups; A_5

A group of prime order is simple (Definition 3.12). To find a non-abelian simple group we need the classes of A_5, since a normal subgroup is a union of classes (Proposition 2.2(ii)). The classes of S_n are its cycle types (Theorem 2.9); in A_n a class of S_n can break in two.

Lemma 4.1 (Classes of A_n). *Let $n \geq 2$, let $\sigma \in A_n$, and let $[\sigma]$ be its class in S_n. Then $[\sigma] \subseteq A_n$, and:*

(i) if $C_{S_n}(\sigma)$ contains an odd permutation, $[\sigma]$ is one class of A_n;

(ii) if $C_{S_n}(\sigma) \leq A_n$, $[\sigma]$ is the union of two classes of A_n, each with $|[\sigma]|/2$ elements.

Proof. The sign is a homomorphism to an abelian group, so

$$\operatorname{sgn}(\tau\sigma\tau^{-1}) = \operatorname{sgn}\sigma = 1.$$

Thus $[\sigma] \subseteq A_n$.

Write $C = C_{S_n}(\sigma)$. The centraliser of σ in A_n is $C \cap A_n$, so by Theorem 1.15(i) the class of σ in A_n has

$$\frac{|A_n|}{|C \cap A_n|} = \frac{n!/2}{|C \cap A_n|}$$

elements. It lies in $[\sigma]$, which has $n!/|C|$ elements.

(i) The sign restricted to C is onto $\{\pm 1\}$, with kernel $C \cap A_n$; so $|C \cap A_n| = |C|/2$ (Theorem 2.3(ii)), and the class of σ in A_n has $n!/|C|$ elements. So it is all of $[\sigma]$.

(ii) Now $C \cap A_n = C$, so the class of σ in A_n has $|[\sigma]|/2$ elements. Let $\sigma' = (1\,2)\,\sigma\,(1\,2)^{-1}$, which exists as $n \geq 2$.

Two classes cover $[\sigma]$. An element of $[\sigma]$ is $\tau\sigma\tau^{-1}$ with $\tau \in S_n$. If τ is even, it lies in the class of σ in A_n. If τ is odd, then $\rho = \tau(1\,2)$ is even, $\tau = \rho\,(1\,2)$, and

$$\begin{aligned}\tau\sigma\tau^{-1} &= \rho\,(1\,2)\,\sigma\,(1\,2)^{-1}\rho^{-1} \\ &= \rho\,\sigma'\rho^{-1},\end{aligned}$$

which lies in the class of σ' in A_n.

The halves. The class of σ holds only half of $[\sigma]$, so the two classes differ; classes of A_n that differ are disjoint, so the class of σ' is the other half. ∎

Example 4.2 (The classes of A_5)**.** The even cycle types in S_5 are (1^5), $(2^2, 1)$, $(3, 1^2)$ and (5) (Theorem 2.10(ii)). For each, Lemma 4.1 asks for an odd permutation that commutes with a representative:

type	σ	$\lvert[\sigma]\rvert = 120/z_\mu$	odd, commuting	classes in A_5
(1^5)	e	1	$(1\,2)$	1
$(2^2, 1)$	$(1\,2)(3\,4)$	15	$(1\,2)$	15
$(3, 1^2)$	$(1\,2\,3)$	20	$(4\,5)$	20
(5)	$(1\,2\,3\,4\,5)$	24	none	12, 12

In the last row the centraliser has order $z_{(5)} = 5$, so it is $\langle(1\,2\,3\,4\,5)\rangle \leq A_5$.

Example 4.3 (Guard: a class of S_n need not be a class of A_n)**.** The 24 five-cycles are one class of S_5 but two of A_5. The odd permutation $(1\,2)$ conjugates $(1\,2\,3\,4\,5)$ to

$$(1\,2)(1\,2\,3\,4\,5)(1\,2)^{-1} = (2\,1\,3\,4\,5),$$

and no even permutation does. Counted by cycle type, A_5 would have four classes, not five; Lecture 7 counts the irreducible representations of a group by its classes.

Theorem 4.4 (A_5 is simple)**.** *The alternating group A_5 is simple.*

Proof. Let $N \trianglelefteq A_5$. Then N is a union of classes of A_5 containing $\{e\}$ (Proposition 2.2(ii)), and $|N|$ divides 60 (Lagrange). By Example 4.2, the sums of 1 and some of 15, 20, 12, 12 are

$$1,\ 13,\ 16,\ 21,\ 25,\ 28,\ 33,\ 36,\ 40,\ 45,\ 48,\ 60,$$

and of these only 1 and 60 divide 60. So $N = \mathbf{1}$ or $N = A_5$; and $A_5 \neq \mathbf{1}$. ∎

Example 4.5 (A_5 is not abelian)**.** Two three-cycles need not commute:

$$\begin{aligned}(1\,2\,3)(3\,4\,5) &= (1\,2\,3\,4\,5), \\ (3\,4\,5)(1\,2\,3) &= (1\,2\,4\,5\,3).\end{aligned}$$

4.2 Composition series and Jordan–Hölder

A proper normal subgroup $N \neq \mathbf{1}$ splits G into smaller groups N and G/N. Repeating gives simple factors. To compare two composition series, we need the second isomorphism theorem.

Theorem 4.6 (Second isomorphism theorem). *Let $H \leq G$ and $N \trianglelefteq G$. Then $HN = \{hn : h \in H,\ n \in N\}$ is a subgroup of G with $N \trianglelefteq HN$; $H \cap N \trianglelefteq H$; and*

$$H/(H \cap N) \cong HN/N, \qquad h(H \cap N) \mapsto hN.$$

Proof. *HN is a subgroup.* It contains e. For $h, h' \in H$ and $n, n' \in N$,

$$(hn)(h'n') = (hh')\,(h'^{-1}nh')\,n',$$
$$(hn)^{-1} = h^{-1}\,(hn^{-1}h^{-1}),$$

which lie in HN, as N is normal in G, hence in HN.

The isomorphism. The quotient map restricted to H, $h \mapsto hN$, is a homomorphism $H \to HN/N$. It is onto, since $(hn)N = hN$, and its kernel is $H \cap N$, normal in H as a kernel. So the first isomorphism theorem (Theorem 2.3(ii)) applies. ∎

If M and N are both normal in G, so is the subgroup MN: for $g \in G$,

$$g\,MN\,g^{-1} = (gMg^{-1})(gNg^{-1}) = MN.$$

Definition 4.7 (Composition series). A *composition series* of G is a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \ldots \trianglerighteq G_r = \mathbf{1}$$

in which each G_{i+1} is a proper normal subgroup of G_i and each *factor* G_i/G_{i+1} is simple; r is its *length*. Two composition series are *equivalent* if they have the same length and their factors agree up to order and isomorphism.

Each G_{i+1} need be normal only in G_i (Example 2.13). The trivial group has one composition series, $\mathbf{1}$, of length 0. A proper normal subgroup N of G is *maximal normal* if no normal subgroup K of G has $N < K < G$.

Lemma 4.8. *For $N \trianglelefteq G$, the quotient G/N is simple if and only if N is a maximal normal subgroup of G.*

Proof. By the correspondence theorem (Theorem 2.3(iii)), the normal subgroups of G/N are the K/N with $N \leq K \trianglelefteq G$; and $G/N \neq \mathbf{1}$ exactly when $N \neq G$. ∎

Theorem 4.9 (Jordan–Hölder). *Let G be a finite group.*

(i) G has a composition series.

(ii) Any two composition series of G are equivalent.

Proof. (i) By induction on $|G|$.

The base. For $G = \mathbf{1}$, take $\mathbf{1}$.

The top factor. If $G \neq \mathbf{1}$, let N be a proper normal subgroup of largest order ($\mathbf{1}$ is one). It is maximal normal, so G/N is simple (Lemma 4.8).

The tail. By induction N has a composition series, and

$$G \trianglerighteq N \trianglerighteq \ldots \trianglerighteq \mathbf{1}$$

is one for G.

(ii) By induction on $|G|$; for $G = \mathbf{1}$ there is only one series. Let $G \neq \mathbf{1}$, with two composition series

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \ldots \trianglerighteq G_r = \mathbf{1},$$
$$G = H_0 \trianglerighteq H_1 \trianglerighteq \ldots \trianglerighteq H_s = \mathbf{1},$$

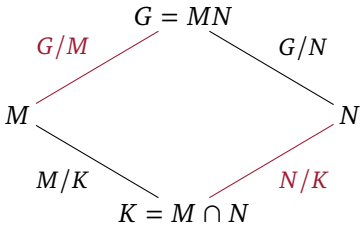
and let $M = G_1$, $N = H_1$, maximal normal by Lemma 4.8.

Case $M = N$. The tails from G_1 and from H_1 are composition series of M, and $|M| < |G|$. By induction they are equivalent; add the common factor G/M.

Case $M \neq N$: the diamond. Then $N \not\leq M$, for otherwise $N < M < G$, against the maximality of N. So MN is normal in G and properly contains M; by the maximality of M, $MN = G$. Let $K = M \cap N$. By Theorem 4.6, K is normal in M and in N, and

$$M/K \cong MN/N = G/N,$$
$$N/K \cong MN/M = G/M,$$

both simple: opposite sides of the diamond are isomorphic.



Case $M \neq N$: the induction. Take a composition series $K = K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1}$ of K, by existence. As M/K is simple,

$$M \trianglerighteq K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1}$$

is a composition series of M, and by induction it is equivalent to $G_1 \trianglerighteq \ldots \trianglerighteq G_r$. So the first series of G is equivalent to

$$G \trianglerighteq M \trianglerighteq K_0 \trianglerighteq \ldots \trianglerighteq K_t = \mathbf{1},$$

with factors G/M, G/N and those of K. Through N, the second has factors G/N, G/M and those of K: the same, in another order. ∎

So the *composition factors* of G, with their multiplicities, depend on G alone.

Example 4.10 (Guard: the factors do not determine the group)**.** The cyclic group $C_6 = \langle g \rangle$ has two composition series,

$$C_6 \trianglerighteq \langle g^2 \rangle \trianglerighteq \mathbf{1}, \qquad C_6 \trianglerighteq \langle g^3 \rangle \trianglerighteq \mathbf{1},$$

with factors C_2, C_3 and C_3, C_2: the order of the factors depends on the series. And $S_3 \trianglerighteq A_3 \trianglerighteq \mathbf{1}$ has the factors C_2, C_3 too, though S_3 is not abelian. Jordan–Hölder fixes the pieces, not how they are put together. Even among abelian groups, C_4 and V_4 differ (only C_4 is cyclic), but both have factors C_2, C_2.

4.3 Solvable groups and the abelianisation

Abelian groups have the simplest pieces. How far is a group from abelian?

Definition 4.11 (Commutators)**.** The *commutator* of $a, b \in G$ is

$$[a, b] = aba^{-1}b^{-1}.$$

The *commutator subgroup* G' is the subgroup generated by all commutators.[a]

[a] Dummit–Foote and Isaacs write $[x, y] = x^{-1}y^{-1}xy$: our $[x^{-1}, y^{-1}]$, a conjugate of $[x, y]$, not in general its inverse. Both generate G'.

So $[a, b] = e$ exactly when $ab = ba$, and $G' = \mathbf{1}$ exactly when G is abelian.

Proposition 4.12 (The abelianisation)**.**

(i) A homomorphism φ carries commutators to commutators, $\varphi([a,b]) = [\varphi(a), \varphi(b)]$; and $G' \trianglelefteq G$.

(ii) For $N \trianglelefteq G$, the quotient G/N is abelian if and only if $G' \leq N$.

(iii) G/G' is abelian, and every homomorphism φ from G to an abelian group factors through it: $\varphi = \bar{\varphi} \circ q$, with $q\colon G \to G/G'$ the quotient map.

Proof. (i) As φ is a homomorphism,

$$\varphi(aba^{-1}b^{-1}) = \varphi(a)\varphi(b)\varphi(a)^{-1}\varphi(b)^{-1}.$$

Conjugation by g is a homomorphism $G \to G$, so it carries commutators to commutators, and $gG'g^{-1} \subseteq G'$, enough for normality (§ 2.1).

(ii) Let $q\colon G \to G/N$ be the quotient map. Homomorphisms preserve commutators, so $q([a,b]) = [q(a), q(b)]$, which is the identity N exactly when $q(a)$ and $q(b)$ commute. So G/N is abelian exactly when every commutator lies in $N = \ker q$, equivalently $G' \leq N$.

(iii) The quotient criterion makes G/G' abelian. If φ maps to an abelian group, then

$$\varphi([a,b]) = [\varphi(a), \varphi(b)] = e,$$

so $G' \leq \ker\varphi$, and φ factors through G/G' by Theorem 2.3(i). ∎

G/G' is the *abelianisation* of G, its largest abelian quotient by the quotient criterion. Factorisation shows that the one-dimensional representations of G, homomorphisms to $\mathbb{C}^\times$, are those of G/G' (Lecture 6).

Example 4.13 (S_3')**.** As S_3/A_3 has order 2, it is abelian, and $S_3' \leq A_3$ by the quotient criterion. With $(1\,3)^{-1} = (1\,3)$ and $(1\,2)^{-1} = (1\,2)$,

$$\begin{aligned}[(1\,3),(1\,2)] &= (1\,3)(1\,2)(1\,3)(1\,2)\\ &= (1\,2\,3)(1\,2\,3) = (1\,3\,2),\end{aligned}$$

which generates A_3. So $S_3' = A_3$, and $A_3' = \mathbf{1}$, as A_3 is abelian.

Definition 4.14 (Solvable group)**.** The *derived series* of G is $G^{(0)} = G$ and $G^{(i+1)} = \left(G^{(i)}\right)'$:

$$G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq G^{(2)} \trianglerighteq \cdots.$$

G is *solvable* if $G^{(m)} = \mathbf{1}$ for some m.

Each term is normal in the one before, with abelian quotient (Proposition 4.12). Abelian groups are solvable, and so is S_3 (Example 4.13). A simple group G that is not abelian is not: G' is normal and not $\mathbf{1}$, so $G^{(i)} = G$ for every i. In particular A_5 is not solvable.

Theorem 4.15 (Solvable groups)**.**

(i) If $H \leq G$, then $H^{(i)} \leq G^{(i)}$ for every i. So subgroups of solvable groups are solvable.

(ii) If $\varphi\colon G \to K$ is a surjective homomorphism, then $\varphi(G^{(i)}) = K^{(i)}$ for every i. So quotients of solvable groups are solvable.

(iii) If $N \trianglelefteq G$, and N and G/N are solvable, then G is solvable.

(iv) G is solvable if and only if it has a chain $G = H_0 \trianglerighteq H_1 \trianglerighteq \ldots \trianglerighteq H_s = \mathbf{1}$, each H_{i+1} normal in H_i, with every H_i/H_{i+1} abelian.

Proof. (i) By induction on i: the commutators of elements of $H^{(i)}$ are commutators of elements of $G^{(i)}$, so $(H^{(i)})' \leq (G^{(i)})'$.

(ii) By induction on i: φ maps the commutators of $G^{(i)}$ onto those of $\varphi(G^{(i)}) = K^{(i)}$ (Proposition 4.12(i)), so it maps $G^{(i+1)}$ onto $K^{(i+1)}$.

(iii) Let $(G/N)^{(n)} = \mathbf{1}$ and $N^{(m)} = \mathbf{1}$. The quotient-map formula gives $G^{(n)} \leq N$. By the subgroup inclusion for derived series,

$$G^{(n+m)} = \left(G^{(n)}\right)^{(m)} \leq N^{(m)} = \mathbf{1}.$$

(iv) The derived series of a solvable group is such a chain. Conversely, $G^{(i)} \leq H_i$ by induction: if $G^{(i)} \leq H_i$, use the subgroup inclusion and Proposition 4.12(ii) in H_i,

$$G^{(i+1)} \leq H_i' \leq H_{i+1}.$$

So $G^{(s)} \leq H_s = \mathbf{1}$. ∎

Corollary 4.16. *S_n is solvable if and only if $n \leq 4$.*

Proof. In $S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1}$ each term is normal in the one before (§ 2.6); the first two factors have prime order, so are cyclic (Corollary 1.12), and V_4 is abelian. By Theorem 4.15(iv), S_4 is solvable. For $n \leq 4$, S_n is the subgroup of S_4 fixing $n+1, \dots, 4$; solvability passes to subgroups.

For $n \geq 5$, S_n contains A_5, as the even permutations of $\{1, \dots, 5\}$. Since subgroups of solvable groups are solvable, S_n is not. ∎

4.4 Nilpotent groups and p-groups

An abelian group is its own centre. A nilpotent group is reached from its centre in finitely many steps.

Definition 4.17 (Nilpotent group)**.** The *upper central series* of G is $Z_0 = \mathbf{1}$ and, for $i \geq 0$, the subgroup $Z_{i+1} \geq Z_i$ with

$$Z_{i+1}/Z_i = Z(G/Z_i).$$

G is *nilpotent* if $Z_m = G$ for some m.

The definition makes sense by induction on i: if $Z_i \trianglelefteq G$, then $Z(G/Z_i)$ is normal in G/Z_i, and by the correspondence theorem (Theorem 2.3(iii)) it is Z_{i+1}/Z_i for exactly one Z_{i+1}, again normal in G. So $Z_1 = Z(G)$, and abelian groups are nilpotent.

Example 4.18 (The group H_1 of order 8)**.** In H_1 (Example 3.3), the first step is the centre,

$$Z_1 = Z(H_1) = \{e,\ (1\,2)(3\,4)\}.$$

The quotient H_1/Z_1 has order 4, so it is abelian (Theorem 3.2(ii)), and it is its own centre. So $Z_2 = H_1$:

$$\mathbf{1} = Z_0 < Z_1 < Z_2 = H_1.$$

H_1 is nilpotent, though not abelian.

Theorem 4.19 (Nilpotent groups)**.**

(i) Every group of order p^a, p prime, is nilpotent.

(ii) Every nilpotent group is solvable.

Proof. (i) Let $|P| = p^a$. If $Z_i \neq P$, then P/Z_i is a non-trivial p-group (Lagrange), so its centre Z_{i+1}/Z_i is not trivial (Theorem 3.2(i)), and

$$|Z_{i+1}| \geq p\ |Z_i|\,.$$

So, for each i, either $Z_i = P$ or $|Z_i| \geq p^i$; and $Z_a = P$.

(ii) Let $Z_m = G$. Each Z_i is normal in G, hence in Z_{i+1}, and each factor $Z_{i+1}/Z_i = Z(G/Z_i)$ is abelian. So the chain

$$G = Z_m \trianglerighteq Z_{m-1} \trianglerighteq \ldots \trianglerighteq Z_0 = \mathbf{1}$$

satisfies Theorem 4.15(iv), and G is solvable. ∎

Example 4.20 (Guard: solvable is not nilpotent). S_3 is solvable (Example 4.13), but $Z(S_3) = \mathbf{1}$ (Example 1.16). So $Z_1 = Z_0 = \mathbf{1}$, and as each Z_{i+1} depends only on Z_i, every Z_i is $\mathbf{1}$: S_3 is not nilpotent. The converse of (ii) fails.

4.5 Worked example

What are the composition series and the derived series of S_4, and what is its abelianisation?

The composition series. The normal subgroups of S_4 are $\mathbf{1}$, V_4, A_4, S_4 (§ 2.6). So A_4 is the only maximal normal subgroup, and every composition series begins $S_4 \trianglerighteq A_4$ (Lemma 4.8).

A normal subgroup of A_4 is a union of its classes, of sizes 1, 3, 4, 4 (Example 3.19), containing $\{e\}$ and of order dividing 12:

$$\{1, 4, 12\}.$$

The union of size 4 is V_4, so the normal subgroups of A_4 are $\mathbf{1}$, V_4, A_4, and the next term is V_4. Its maximal normal subgroups are its three subgroups of order 2. So S_4 has exactly three composition series,

$$S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \langle v \rangle \trianglerighteq \mathbf{1},$$

one for each $v \in V_4$ with $v \neq e$. The factors have orders 2, 3, 2, 2, so they are C_2, C_3, C_2, C_2 (Corollary 1.12), the same for all three, as Theorem 4.9 says.

The derived series. As S_4/A_4 is abelian, $S_4' \leq A_4$. It contains the 3-cycle $[(1\,3), (1\,2)]$ of Example 4.13, and it is normal (Proposition 4.12). Of $\mathbf{1}$, V_4, A_4, only A_4 contains a 3-cycle; so $S_4' = A_4$.

Next, A_4/V_4 has order 3, so it is abelian, and $A_4' \leq V_4$. It contains

$$\begin{aligned}[(1\,2\,3), (1\,2)(3\,4)] &= (1\,2\,3)\,(1\,2)(3\,4)\,(1\,2\,3)^{-1} \cdot (1\,2)(3\,4) \\ &= (1\,4)(2\,3) \cdot (1\,2)(3\,4) = (1\,3)(2\,4),\end{aligned}$$

so it is not $\mathbf{1}$; it is normal in A_4, so $A_4' = V_4$. Finally $V_4' = \mathbf{1}$, as V_4 is abelian. The derived series is

$$S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1},$$

with factors C_2, C_3 and $V_4 \cong C_2 \times C_2$. It is shorter than a composition series: its last factor is abelian but not simple.

The abelianisation.

$$S_4/S_4' = S_4/A_4 \cong C_2.$$

By Proposition 4.12(iii), every homomorphism from S_4 to an abelian group factors through the sign.

Not nilpotent. Only e has a class of one element in S_4 (§ 2.6), so $Z(S_4) = \mathbf{1}$, and, as for S_3, every Z_i is $\mathbf{1}$. So S_4 is solvable but not nilpotent.

Self-check. Is A_4 solvable? Is it nilpotent? *Answer:* solvable, with derived series $A_4 \trianglerighteq V_4 \trianglerighteq \mathbf{1}$. Not nilpotent: its classes have 1, 3, 4, 4 elements, so $Z(A_4) = \mathbf{1}$, and every Z_i is $\mathbf{1}$.

Summary.

- *The main results:* A_5 is simple (Theorem 4.4); the composition factors of a finite group are unique (Theorem 4.9); solvability passes to subgroups, quotients and extensions (Theorem 4.15), and S_n is solvable exactly when $n \leq 4$ (Corollary 4.16); p-groups are nilpotent, and nilpotent groups solvable (Theorem 4.19).
- *Handed on:* G', with G/N abelian exactly when $G' \leq N$ (Proposition 4.12), to Lecture 6's one-dimensional representations; solvability and Theorem 4.15(iii) to Lecture 9's $p^a q^b$ theorem.
- *Next:* Lecture 5 begins the representations: when can a linear action of a group be split into irreducible pieces?

References

Dummit and Foote, *Abstract Algebra*: the second isomorphism theorem, §3.3, Thm 18, p. 97; simple groups and composition series, §3.4, pp. 102–103, Jordan–Hölder, Thm 22, p. 103, its proof outlined in Exercises 9–10, p. 106; solvable groups and extensions, p. 105; A_5 simple, §4.3, Thm 12, p. 128, and the splitting of classes, Exercises 19–20, p. 131; commutators, §5.4, Prop. 7, pp. 169–170; the upper central series, §6.1, p. 190, and p-groups nilpotent, Prop. 2, p. 191; the derived series, Thm 9 and Prop. 10, pp. 195–196. Alperin and Bell, *Groups and Representations*: Jordan–Hölder by two maximal normal subgroups, §10, pp. 91–92. Rotman, *An Introduction to the Theory of Groups*: commutators as here, and G/H abelian iff $G' \leq H$, Thm 2.23, p. 33.